SEMESTER - VIII

COURSE: MAJOR – 19 (MJ-19)
PAPER NAME: NUCLEAR AND PARTICLE PHYSICS

TOTAL CREDITS: THEORY-03, PRACTICAL-01
TEACHING HOURS: THEORY-45, PRACTICAL-30

EVALUATION			
	External Exam	**Internal Exam**	**Practical**
Full Marks	**60**	15 (10 Written + 5 Attendance/Overall Class Performance)	**25**
Duration of Exam	3 Hours	1 Hour	6 Hours
Pass Marks	30 Marks		10 marks

Course Objective:

The course on Nuclear Physics aims to provide students with a comprehensive understanding of nuclear forces, reactions, properties of nuclei, radioactivity decay, accelerators and detectors, as well as an introduction to particle physics. Students will gain knowledge about the fundamental principles governing nuclear interactions and the behavior of subatomic particles.

Learning Outcomes:

1. Understand nuclear forces, scattering processes, nuclear reactions, and their conservation laws.
2. Analyze nuclear properties, radioactivity types, decay mechanisms, and nuclear stability concepts.
3. Explore nuclear models including liquid drop, shell model, and nuclear magic numbers.
4. Examine particle physics fundamentals, quantum numbers, particle classification, and the quark model.
5. Gain insight into nuclear fission, fusion, reactors, and modern particle physics including the Standard Model and beyond.

PART 'A'

THEORY COURSE CONTENTS:

UNIT-I:

Nuclear Forces and Reactions (15 Lectures): Ground state of Deuteron, Neutron-Proton Scattering at low energies, Proton-Proton Scattering at low energies, Analysis of n-p and p-p scattering, Interpretation of p-p and n-n scattering. Types of Nuclear Reactions, Conservation Laws, kinematics of reactions, Q-value,

Reaction rate, Reaction cross section, Concept of Compound and Direct Reaction, Resonance reaction, Statistical theory of Nuclear reactions, Optical model of nuclear reaction at low energies, Coulomb scattering (Rutherford scattering), Nuclear Fission. Nuclear reactors, Nuclear Fusion, Nuclear Fusion in stars, Fusion reactors.

UNIT-II:

General Properties of Nuclei and Nuclear Models (20 Lectures): Scattering of alpha particles, Constituents of nucleus and their Intrinsic properties, quantitative facts about mass, radii, charge density (matter density), binding energy, average binding energy and its variation with mass number, main features of binding energy versus mass number curve, N/A plot, angular momentum, parity, magnetic moment, electric moments, nuclear excites states. Nuclear stability and Radioactivity, Activities and half-life, Types of radioactivity- (a) Alpha decay: basics of -decay processes, theory of - emission, Gamow factor, Geiger Nuttall law, α-decay spectroscopy, (b) β-decay: energy kinematics for β-decay, positron emission, electron capture, neutrino hypothesis, (c) Gamma decay: Gamma rays emission & kinematics, internal conversion. **Nuclear Models:** Liquid drop model approach, Semi-empirical mass formula and significance of its various terms, condition of nuclear stability, two nucleon separation energies, Fermi gas model (degenerate fermion gas, nuclear symmetry potential in Fermi gas), evidence for Nuclear Shell structure, Nuclear magic numbers, Basic assumption of Shell model, concept of mean field, residual interaction, concept of nuclear force, L-S coupling, j-j coupling, Transformation between L-S and j-j coupling schemes.

UNIT-III:

Particle Physics (10 Lectures): Four fundamental interactions. Quantum numbers -- spin, isospin, strangeness, parity, hypercharge, Conservation laws, Particle classification -- hadron and lepton, Quark model of hadron -- baryon and meson, Gell-Mann plot, Elementary discussion of key experiments that led to the current understanding of unified electro-weak interaction and strong interaction, Standard Model, Elementary exposition of diagrammatic techniques (without actual calculation) used to evaluate cross-sections of production processes and decay rates, Introduction to physics beyond the Standard Model.

SUGGESTED READINGS:

1. Introductory nuclear Physics by Kenneth S. Krane (Wiley India Pvt. Ltd., 2008).
2. Nuclear Physics by D. C. Tayal, Himalaya Publishing House.
3. Nuclear Physics by S. N. Ghoshal, S. Chand & Company.
4. Concepts of nuclear physics by Bernard L. Cohen. (Tata Mcgraw Hill, 1998).
5. Quarks and Leptons, F. Halzen and A.D. Martin, Wiley India, New Delhi.

6. Basic ideas and concepts in Nuclear Physics - An Introductory Approach by K. Heyde (IOP- Institute of Physics Publishing, 2004).

7. Radiation detection and measurement, G.F. Knoll (John Wiley & Sons, 2000).

8. Physics and Engineering of Radiation Detection, Syed Naeem Ahmed (Academic Press, Elsevier, 2007).

PART 'B'

PRACTICAL COURSE CONTENTS:

List of Practical:

1. To determine the half-life of a radioactive isotope (using virtual lab or by simulation).
2. To verify that the intensity of gamma radiation decreases with square of the distance (use virtual lab or by simulation).
3. Study of attenuation of beta or gamma radiation in lead/aluminum and find absorption coefficient (use virtual lab or by simulation).
4. Compute the binding energy, magnetic moment, and estimate the size of the deuteron using simple potential models.
5. Simulate Coulomb (Rutherford) scattering and plot the differential cross-section as a function of angle.
6. Determine nuclear radius from alpha particle scattering data and plot nuclear radius as a function of mass number (A).
7. Plot binding energy per nucleon versus mass number using empirical data and interpret stability features.
8. Verify exponential decay law using simulated data for alpha, beta, and gamma decay and extract half-lives.

SUGGESTED READINGS:

1. https://phet.colorado.edu/
2. Advanced Practical Physics for students, B.L. Flint and H.T. Worsnop, 1971, Asia Publishing House
3. A Text Book of Practical Physics, I. Prakash& Ramakrishna, 11th Ed., 2011, KitabMahal
4. Advanced level Physics Practicals, Michael Nelson and Jon M. Ogborn, 4th Edition, reprinted 1985, Heinemann Educational Publishers

5. A Laboratory Manual of Physics for undergraduate classes, D.P. Khandelwal, 1985, Vani Pub.

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SEMESTER - VIII

COURSE: MAJOR – 20(MJ-20) **TOTAL CREDITS:** THEORY-03, PRACTICAL-01
PAPER NAME: DISSERTATION **TEACHING HOURS:** THEORY-45, PRACTICAL-30

EVALUATION			
	External Exam	**Internal Exam**	**Practical**
Full Marks	**60**	15 (10 Written + 5 Attendance/Overall Class Performance)	**25**
Duration of Exam	3 Hours	1 Hour	6 Hours
Pass Marks	30 Marks		10 marks

Course Objectives:

Enhance students' communication skills through active participation in seminars and group discussions. Prepare them for academic and professional environments where effective communication and teamwork are vital for success. Foster confidence, clarity, and coherence in expressing ideas, enabling students to engage meaningfully in collaborative and interdisciplinary settings.

Course Outcomes:

1. Develop effective verbal and non-verbal communication skills through seminars and group discussions.
2. Demonstrate confidence, clarity, and coherence in presenting and articulating ideas.
3. Collaborate successfully in academic and professional settings, applying interpersonal skills and teamwork strategies in diverse and interdisciplinary environments.

PART 'A'

Course Contents:

The Head of the Department and faculty members will assign topic to the students from the course content of semester VII and VIII. Students will have to work under the supervision of a teacher of the department. Each and every student has to submit electronically typed hardbound dissertation along with the raw data on a week before the examination.

Evaluation Process: Presentation and the report submitted by the students will be evaluated by one external member and one internal member. The External Member may be a Permanent faculty members working in the postgraduate department of the university or other colleges or Retired Professor/Associate Professor/Assistant Professor of the university.

Distribution of marks:

Dissertation Report	25 marks
Presentation	50 marks
Subject Knowledge	25 marks

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SEMESTER - VIII

COURSE: ADVANCE MAJOR – 2 (AMJ-2) **TOTAL CREDITS:** THEORY-03, PRACTICAL-01
PAPER NAME: CLASSICAL DYNAMICS **TEACHING HOURS:** THEORY-45, PRACTICAL-30

EVALUATION			
	External Exam	**Internal Exam**	**Practical**
Full Marks	**60**	15 (10 Written + 5 Attendance/Overall Class Performance)	**25**
Duration of Exam	3 Hours	1 Hour	6 Hours
Pass Marks	30 Marks		10 marks

Course Objective:

This course aims to develop a foundational understanding of small amplitude oscillations, fluid dynamics, and special relativity. Students will explore normal modes, fluid flow principles, and relativistic concepts including Lorentz transformations and four-vectors, enabling them to analyze classical and relativistic physical systems using mathematical and conceptual tools.

Learning Outcomes:

1. Understand stable equilibrium and analyze small amplitude oscillations using normal modes.
2. Apply matrix methods to determine eigen-frequencies and eigen-vectors in coupled oscillatory systems.
3. Explain fluid properties, continuity equation, Poiseuille's law, and fluid-electric current analogies.
4. Solve problems involving Navier-Stokes equation and interpret flow types using Reynolds number.
5. Comprehend special relativity concepts, Lorentz transformations, four-vectors, and their physical significance.

PART 'A'

THEORY COURSE CONTENTS:

UNIT-I:

Small Amplitude Oscillations (15 Lectures): Minima of potential energy and points of stable equilibrium, small amplitude oscillations about the minimum, normal modes of longitudinal simple harmonic oscillations

(maximum 3 masses connected by 4 springs). Kinetic energy (T) and potential energy (V) in terms of normal co-ordinates. T and V matrices: finding eigen-frequencies and eigen-vectors using these matrices.

UNIT-II:

Fluid Dynamics (10 Lectures): Density ρ and pressure P in a fluid, an element of fluid and its velocity, continuity equation and mass conservation, stream-lined motion, laminar flow, Poiseuille's equation for flow of a liquid through a pipe. Analogy between liquid flow and current flow, rate of liquid flow through capillaries in series and in parallel combination. Navier Stokes equation, Reynolds number.

UNIT-III:

Special Theory of Relativity (20 Lectures): Postulates of Special Theory of Relativity, Lorentz Transformations, Consequences of Lorentz Transformations: length contraction, Simultaneity, Time dilation, Addition of velocities, Minkowski space, The invariant interval, light cone and world lines, Space- time diagrams. **Four-vectors**: space-like, time-like and light-like, Examples of Four-vectors: Position Four-vectors, Velocity Four-vectors, Momentum Four-vectors, Acceleration Four-vectors, and Four-force Minkowski force, Doppler Effect from a four-vector perspective, Conservation of Four-momentum, Geometrical Interpretation of Lorentz Transformations, Simultaneity, Length Contraction and Time Dilation.

SUGGESTED READINGS:

1. Boylestad R. L. and Nashelsky L., Electronic Devices and Circuit Theory, Pearson.
2. RaychaudhuriBarun, Electronics: Analog and Digital, Cambridge University Press.
3. Cathey J. J., Schaum's Outline of Theory and Problems of Electronic Devices and Circuits, McGraw-Hill.
4. Helfrick A. D. and Cooper W. D., Modern Electronic Instrumentation and Measurement Techniques, PHI.
5. Malvino A. P. and Bates D. J., Electronic Principles, McGraw-Hill Education.
6. Millman J. and Halkias C. C., Integrated Electronics: Analog and Digital Circuits and Systems, McGraw-Hill, Inc.
7. Streetman B. G. and Banerjee S.K., Solid State Electronic Devices, PHI.
8. Gayakwad R. A., Op-Amps and Linear Integrated Circuits, Pearson.

PART 'B'

PRACTICAL COURSE CONTENTS:

List of Practical:

1. Compute and visualize time-like, space-like, and light-like intervals and vector magnitudes in spacetime.
2. Compute the frequency of small oscillations about a stable minimum of a given potential (e.g., quadratic, anharmonic).
3. Simulate the normal modes and frequencies of longitudinal oscillations of a system of 2 masses and 3 springs.
4. Demonstrate conservation of mass in fluid flow using simulated fluid streamlines.
5. Compute time and position transformations between two inertial frames using Lorentz transformation equations.
6. Simulate the four-velocity and four-acceleration for uniformly moving and accelerating particles.

SUGGESTED READINGS:

1. https://phet.colorado.edu/
2. Advanced Practical Physics for students, B.L. Flint and H.T. Worsnop, 1971, Asia Publishing House
3. A Text Book of Practical Physics, I. Prakash& Ramakrishna, 11th Ed., 2011, KitabMahal
4. Advanced level Physics Practicals, Michael Nelson and Jon M. Ogborn, 4th Edition, reprinted 1985, Heinemann Educational Publishers
5. A Laboratory Manual of Physics for undergraduate classes, D.P. Khandelwal, 1985, Vani Pub.

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SEMESTER - VIII

COURSE: ADVANCE MAJOR – 3(AMJ-3) **TOTAL CREDITS:** THEORY-03, PRACTICAL-01
PAPER NAME: ADVANCED STATISTICAL MECHANICS **TEACHING HOURS:** THEORY-45, PRACTICAL-30

EVALUATION			
	External Exam	**Internal Exam**	**Practical**
Full Marks	**60**	15 (10 Written + 5 Attendance/Overall Class Performance)	**25**
Duration of Exam	3 Hours	1 Hour	6 Hours
Pass Marks	30 Marks		10 marks

Course Objective:

This course aims to develop a comprehensive understanding of the principles and mathematical foundations of thermodynamics, statistical mechanics, and critical phenomena. It introduces students to classical and modern approaches for analyzing many-body systems, including interactions, fluctuations, phase transitions, and time-dependent processes.

Learning Outcomes:

1. Apply thermodynamic principles and Maxwell relations to physical systems like ideal gases, magnetic, and dielectric materials.
2. Analyze interacting many-body systems using cluster expansion, correlation functions, and derive equations of state.
3. Explain the theory of phase transitions and critical phenomena using models like the Ising model, Landau theory, and mean-field approximations.
4. Use renormalization group methods to understand scaling, universality, and critical exponents in low-dimensional systems.
5. Describe time-dependent phenomena such as Brownian motion, linear response, and fluctuation-dissipation relations using Langevin and Kubo formalisms.

PART 'A'

THEORY COURSE CONTENTS:

UNIT-I:

Review of Thermodynamics (15 Lectures): Extensive and intensive variables, Laws of thermodynamics, Legendre transformations and thermodynamic potentials, Maxwell relations, Applications of thermodynamics to (a) ideal gas, (b) magnetic material, and (c) dielectric material, Binomial, Poisson and Gaussian distributions, Central limit theorem, Cluster expansions, Equation of State, Virial coefficients, Pair correlation function and thermodynamic properties, Structure factor, Ornstein-Zernike equation, Debye-Hueckel theory.

UNIT-II:

Phase transitions and critical phenomena (15 Lectures): Gas-liquid and paramagnetic-ferromagnetic transitions, Ising model, Equivalence with other models, Spin-spin correlation function, Mean-field theory, Landau theory. Phase transitions and thermodynamic functions, Second-order transition and critical exponents, Mean field theories, Fluctuations, Upper and lower critical dimensions, Scaling and universality, Renormalization group formulation, Flows, Fixed points, Examples of 1D and 2D Ising systems, Migdal-Kadanoff procedure, Momentum shell renormalization group, The Gaussian model.

UNIT-III:

Time dependent phenomena(15 Lectures): Dynamic correlation and response functions, Symmetry and dispersion relations, Brownian motion and Langevin equations, Velocity auto-correlation and self-diffusion, Linear response theory, Dissipation, The fluctuation-dissipation theorem, The Kubo formula, Onsager's postulate.

SUGGESTED READINGS:

1. Statistical Mechanics: R. K. Pathria (Elsevier).
2. Statistical Mechanics: L. D. Landau and E. M. Lifshitz (Pergamon).
3. Equilibrium Statistical Physics: M. Plischke and B. Bergerson (World Scientific).
4. Principles of Condensed Matter Physics: P. M. Chikin and T. C. Lubensky (Cambridge University Press).
5. Statistical Physics of Fields: M. Kardar (Cambridge University Press).
6. Statistical Physics-I and -II: Kubo, Toda and Ashitsume (Springer).

PART 'B'

PRACTICAL COURSE CONTENTS:

List of Practical:

1. Simulate P-V-T behavior of an ideal gas and validate the ideal gas law across different thermodynamic paths.
2. Compute virial coefficients from model interaction potentials using low-order cluster expansions.
3. Simulate the random walk of a Brownian particle and compute mean square displacement over time.
4. Simulate the Ising model and observe the phase transition and magnetization curves.
5. Compute and plot the spin-spin correlation function as a function of distance and temperature in the Ising model.
6. Apply the Kubo formula to compute conductivity or susceptibility from correlation functions.

SUGGESTED READINGS:

1. https://phet.colorado.edu/
2. Advanced Practical Physics for students, B.L. Flint and H.T. Worsnop, 1971, Asia Publishing House
3. A Text Book of Practical Physics, I. Prakash& Ramakrishna, 11th Ed., 2011, KitabMahal
4. Advanced level Physics Practicals, Michael Nelson and Jon M. Ogborn, 4th Edition, reprinted 1985, Heinemann Educational Publishers
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SEMESTER - VIII

COURSE: RESEARCH PROJECT/DISSERTATION – 1(RP/D-1) **TOTAL CREDITS:** 08
PAPER NAME: RESEARCH PROJECT OR DISSERTATION

Note: Students who secure 75% marks and above in the first six semesters and wish to undertake research at the undergraduate level can choose a research stream in the fourth year.

Course Objectives:

The objectives of the course is to facilitate students to carry out extensive research and develop as self-guided learning and analytical skills through problem and gap identification, development of research methodology, interpretation of findings and presentation of results.

Learning Outcomes: After completion of the course, the learners will be able to:

1. Gain in-depth knowledge in the major field of study.
2. Design and justify research methodology.
3. Utilize appropriate research methodology for data collection
4. Analyze the collected data and draws conclusions accordingly.

Course Contents:

The HOD of the department has to allot supervisor to the students from among the faculty members who have PhD degree. After that the students have to select a research problem with the help of the supervisor and they have to submit a summary or research proposal to the department. Thereafter, the HOD of the department will organize a meeting of the Departmental Committee and after the presentation of the student the committee will approve or reject his/her synopsis/research proposal. Students will start their research work after getting approval from the departmental committee.

At the end of the semester the student has to submit the project dissertation to the department and that will be evaluated by the following members:

(i) HOD of the Department–Chairmen

(ii) HOD, University Department/ Nominated Faculty - External member

(ii) Faculty members of the department – Internal member

External members may be any of the following:

1. Permanent professors working in the postgraduate department of the university or other colleges who have the qualification to become PhD supervisors.

OR

2. Retired Professor/Associate Professor/Assistant Professor of the university who has been supervising PhD scholar.

OR

3. Professor/Associate Professor/Assistant Professor of the outside university who has been supervise PhD scholar.

Note- The project dissertation will be evaluated under the following heads:

1. Motivation for the choice of topic
2. Project dissertation design
3. Methodology and Content depth
4. Results and Discussion
5. Future Scope & References
6. Participation in Internship programme with reputed organization
7. Application of Research technique in Data collection
8. Report Presentation
9. Presentation style Broad

Guidelines for distribution of marks may be as follows or as appropriate:

Assessment of project synopsis	75 marks
Assessment of project Thesis	100 marks
Viva-voce	25 marks

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